Benha faculty of Computers
and Artificial Intelligence

DRIVER MONITERING SYSTEM (DMS)

Supervised by : Dr / Mostafa Abdelsallam

Presented by :

- Ahmed Abdelaal Ahmed Abo-Hadeed.
- Ali Nabil Ali .
- idris tarek El-Sayed.
- Elsayed Sameh Elsayed.
- Ahmed Mohmed Gouda.
- Rania Yasser Mohamed.
- Sherine Ebrahim.
- Eman Hamada Abdelhady .
- Khloud Elsayed Elsayed.

Abstract

Road accidents are very common all over the world. It is due to the lack of attention of drivers. Facts on traffic accidents state that driver's mistake is the major reason of loss and harm on roads all over the world every day. In this Project we described a module for intelligent driver monitoring system to decrease the extent of such losses which can automatically detects the driver distraction. As the distracted driving has been concerned as a causal aspect in many accidents, therefore a real-time driver monitoring system can prevent traffic accidents effectively. This project expresses the techniques to monitor driver safety by evaluating information associated to the visual features of the driver. This monitoring system normally works based on driving prototype, driver's video and images

1.Introduction

According to the data information, more than 1.2 million people die every year due to road accidents and more than 20 million people experienced non-serious injuries due to road accidents [1][20] . For this purpose, the development of driver monitoring system has become very important. This system mostly identifies the driver's distraction level by the interval of head movement and head swing. We present a system for analysing driver visual attention. The method is based on the on estimation of the overall figures to accurately detect a person's head and facial features. The system classifies rotation in all viewing directions. In addition; it is able to detect the eye closure as well as head movement. The system accurately detects the drowsiness and distraction by using the accurate measure of algorithms and it can be used for more advanced driver visual attention monitoring. The system has been tested with different driving habits and with different users. In order to avoid these destructive accidents, the status of drowsiness of the driver should be monitored. The behavioural measures has been used extensively for monitoring drowsiness like eye closure, yawning, head pose and head movement is monitored through a particular camera fixed in the car in front of the driver and the driver is alerted or warned if any of these actions of driver are detected.

2.Objectives

- **Enhance Driver Safety**: Implement a DMS to significantly reduce the occurrence of accidents and incidents caused by driver fatigue, distraction, or other risky behaviors.

- **Reduce Accident Frequency**: Decrease the number of accidents involving our fleet vehicles by [specify a measurable percentage], compared to the baseline data before DMS implementation.

- **Minimize Accident Severity**: Reduce the severity of accidents, as measured by the cost of repairs, medical expenses, and downtime, by [specify a measurable percentage] after the DMS is operational.

- **Improve Driver Alertness**: Enhance driver alertness and attentiveness by [specify a measurable improvement metric] through timely alerts and interventions provided by the DMS.

- **Ensure Regulatory Compliance**: Ensure that all drivers and vehicles are compliant with regional and industry-specific regulations related to driver monitoring, safety, and data privacy.

- **Provide Data-Driven Insights**: Leverage the data collected by the DMS to establish a comprehensive analytics framework that informs decision-making, enabling continuous improvement in driver safety and operational efficiency.

- **Implement Driver Training Programs**: Develop customized driver training programs based on DMS data to address specific areas of improvement and enhance driver skills.

- **Enhance Safety Culture**: Foster a culture of safety within the organization, emphasizing the importance of responsible and alert driving practices among all drivers.

- These objectives collectively form a comprehensive roadmap for the development and implementation of the Driver Monitoring System. They are designed to measurably enhance driver safety, reduce accidents and associated costs technological adaptability.

3.Scope

there are many products out there that provide the detection of drivers' drowsiness which are implemented in many vehicles. This driver drowsiness detection system provides the functionalities that may seem similar to the previous work, with better results and additional features. The system aims to not only detect the driver's drowsiness but also measure the driver's drowsiness level. In addition, the system has a specific action to make with respect to the drowsiness level of the driver with a suitable alert.

3.1.Features

- System detects the drowsiness level of the driver
- Specific action is taken according to the driver's drowsiness level
- Actions taken are accompanied with a suitable alarm system with respect to the drowsiness level
- Driver can respond to the alert through a responsive touch screen
- Emergency message is sent with location in case of high-risk drowsiness to emergency contacts and ambulance.
- If the system detect Activates that distract the driver like Calling , drinking ,Smoking or holding his Phone the system sends an alert to the driver to make him more focused .
- The driver can make emergency call to the emergency contact saved in the system.

4.2.Inclusions

- **Real-Time Monitoring:** The DMS will continuously monitor driver behavior in real-time, including but not limited to eye movement, head positioning, and steering patterns.
- **Alerting Mechanisms:** The system will provide immediate alerts to drivers for potential risks such as drowsiness, distraction, or aggressive driving through auditory, visual, or haptic feedback.
- **Data Collection:** The DMS will collect and store relevant driver behavior data, including timestamps, alert occurrences, and contextual information, for later analysis.
- **Data Analysis:** Data collected by the DMS will be subject to analysis to identify patterns, trends, and areas for improvement in driver safety.
- **Reporting:** The system will generate regular and ad-hoc reports summarizing driver performance, safety incidents, and recommendations for corrective actions.
- **Driver Identification:** The DMS will include driver identification capabilities to link recorded data to specific drivers.
- **Compliance:** The system will be designed to ensure compliance with all relevant legal and industry-specific regulations related to driver monitoring, data privacy, and safety.
- **Training Integration:** Data insights from the DMS will be integrated into driver training programs to address specific areas of improvement.

4-Methodology

1. After the system start **, The camera** starts taking frames and send it to the SOC like Raspberry pi .
2. The Raspberry pi makes the processing using AI techniques:
   - Computer vision like open cv , object detection , MediaPipe and other techniques ... .
   - Machine learning like KNN ,Support Vector Machines (SVM) ,and other techniques... .
   - Neural Network and Deep Learning like CNN and RNN

to detect:

- Verifying the driver's identity .
- Driver Drowsiness levels .
- Drivers Expression if he is nervous or on mode that doesn't make him Naturel the system send alert .
- And detect what is the action Driver do if he Smoking or drinking or eating the System sent alarm and suggest to Take A break until he finishing from his action.
- And system can Detect Driver Eye Blink ,Head position and Zone ,Eye Gaze Direction and Zone .

1. after detect these techniques we can detect the level of Drowsiness and The driver's concentration while driving
2. if is the high level of Drowsiness and Distraction the driver send alert on screen and make high alarm using sound system

4.1.Scenarios

The system has multiple scenarios that are handled according to the driver’s drowsiness level:

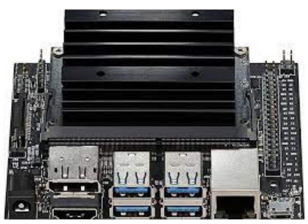
1. The first scenario is the normal scenario; the user is not drowsy so no action should be taken, and the clock is displayed on the screen.
2. The second scenario is when the risk level of the driver drowsiness is low and the driver making some activities like eating or drinking . In this case, a gentle alarm must be activated to notify the driver with a warning on screen. There is no need for the driver to respond to this warning.
3. The third scenario is when the risk level of the driver drowsiness is high. In this case, show a message that alerts him to which he must responds. If no response in 10 seconds, then the waiting lights should be activated, and the car self-drive and self-park on the right side. After self-parking, a very high tone alarm is activated, and an SOS message is sent to emergency contacts and ambulance. If response is given before 10 seconds, then the driver can take control again.

4.2.ProposedHardware Components

- **A System-on-a-Chip (SoC) board:** is the core of the hardware components since every hardware component is connected to it .used for connecting to other components in addition to several other pins and ports such as the HDMI port for display. It supports high-resolution sensors and can process several sensors in parallel. Ex :- Raspberry pi Model B or NVIDIA Jetson Nano

Raspberry pi Model B

NVIDIA Jetson Nano



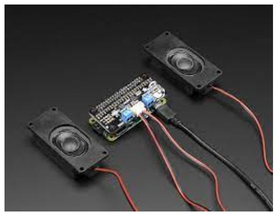
4-Methodology

- **Camera:** the camera used for detecting the frames of the driver and send it to the board to make the processing .   ex:Raspberry pi Camera

Raspberry pi Camera

Screen

- **Screen:** To communicate with the driver behind the wheels



- **Speakers:** The driver is alerted in another way other than a message on the screen, which is a sound played on the speakers

- **SIM 808 GPS/GSM Module:** Since the driver could be already not responding to the alert messages on the screen, a fast action is needed to be taken. To minimize the damage and to prevent the loss of innocent lives, an SOS signal is sent to the emergency operator through the modem. The driver can make an emergency call.

- **USB to TTL Module:** he USB to TTL Module is a converter that is used to convert the serial ports into a single USB 2.0 type-A in order to be used by computers that do not have serial ports